Simple visualisation techniques for simplified Humanities. A survey of Digital Humanities projects

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Introduction

Generous interfaces (Whitelaw 2015) and storytelling tools (Renda et al. 2023) are argued to support interpretation in Cultural Heritage, yet humanistic knowledge cannot be easily embedded into visualisation systems (Drucker 2011; 2014). Scholars advocate for visualisations exposing interpretative aspects (Drucker 2011; 2014; Windhager et al. 2019; Trocchianesi / Bollini 2023; Elli et al. 2021; Viola 2023), encouraging slow visualisation (Elli et al. 2021; Hinrichs et al. 2019; Bradley et al. 2016) and epistemic potential (Münster / Terras 2020; Schwan et al. 2019).

Domain-agnostic taxonomies and classifications describe interaction techniques (Shneiderman 1996; Yi et al. 2007; Dimara / Perin 2020; Brehmer / Munzner 2013) and narrative design spaces (Segel / Heer 2010; McKenna et al. 2017; Zhao / Elmqvist 2023). Though uncertainty visualisation taxonomies (Kinkeldey et al. 2014) informed DH research (Panagiotidou et al. 2023), surveys reveal limited attempts to visualise uncertainty, scarce narrativity, and use of multiple visualisations (Khulusi et al. 2020; Windhager et al. 2019; Panagiotidou et al. 2023). However, studies remain narrow-scoped, disregarding approaches to humanities enquiry. This work addresses uncertainty in DH projects and introduces novel classificatory aspects for tailored visualisations.

Methods

We analysed 186 web-based DH projects collected from the literature (Windhager et al. 2019) and follow-up projects (33 projects), DH associations' websites (AIUCD[1], EADH[2]) and research centres (104 projects), and the Digital Humanities Awards website[3] (49 projects). Selected entities include accessible web projects without access barriers that significantly rely on visualisations.

Our iterative methodology, inspired by (Segel / Heer 2010), developed a classification schema through direct observation and interaction with the visualisation systems, minimising bias from discrepancies between documented features and actual implementation. The framework adapts previous models, covering: 1) narrativity and narrative strategies (McKenna et al. 2017); 2) humanities domains (Terras at al. 2016: 138; Schreibman et al. 2004); 3) visualisation techniques (Windhager et al. 2019); 4) interaction techniques (Yi et al. 2007; Dimara / Perin 2020; Brehmer / Munzner 2013); 5) uncertainty visualisation methods (Kinkeldey et al. 2014; Panagiotidou et al. 2023; Drucker 2011); and 6) critical adaptations, identifying tailored visualisations that embrace humanities-specific complexity, promoting slow looking inquiry.

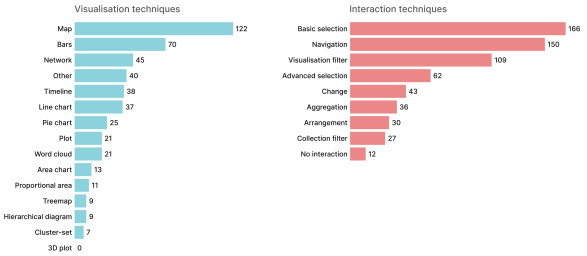
Three research questions guided our analysis:

- **RQ1.** How do DH projects employ information visualisation, interaction, and narrative techniques?
- **RQ2.** How do DH projects address humanities visualisation needs?
- **RQ3.** How do DH projects differ from projects in other humanities-related contexts?

Results and discussion

The dataset (Battisti 2024) is analysed in a dedicated Jupyter Notebook (Battisti 2025).

Our findings confirm a naive reuse of visualisation techniques. DH projects use more maps and fewer networks compared to previous studies (Windhager et al. 2019; Khulusi et al. 2020; Panagiotidou et al. 2023), likely due to network readability issues (Koponen / Hildén 2019). Interactive strategies are limited to easy-to-deploy options, with some non-interactive visualisations present (Figure 1). Specific visualisation and interaction patterns were identified using the Apriori algorithm[4], while a small positive correlation exists between the number of visual and interaction techniques (Spearman coefficient=0.394; p-value=2.54e-08).

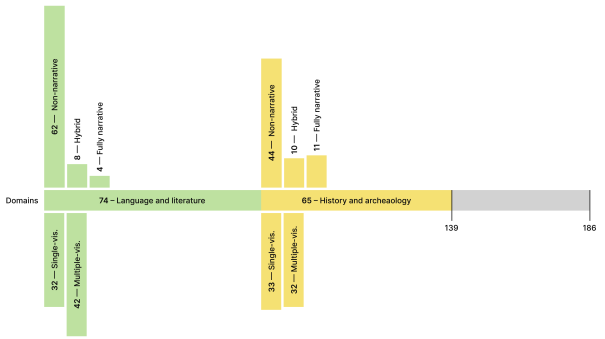


Frequency of visualisation and interaction techniques.

Multiple visualisations are less used than in Cultural Heritage collections (Windhager et al. 2019). Narrative strategies remain underutilised despite growing advocacy (Gershon / Page 2001; Renda et al. 2023). When present, narratives are linear and author-driven, often using tools like StoryMapJS[5] and TimelineJS[6] with minimal customisation (Figure 2).

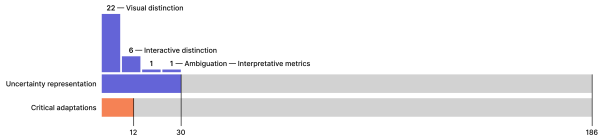
Distribution of narratives and decomposition in narrative progression (top) and type (bottom).

“Language and literature” and “history and archaeology” are the most represented domains, each characterised by narrative strategies and visualisations usage (Figure 3).



Narrativity (top) and multi-visualisation approaches (bottom) in "language and literature" and "history and archaeology" domains.

Few DH projects critically adapt visual solutions, with uncertainty visualisation remaining underrepresented (Khulusi et al. 2020; Panagiotidou et al. 2023) (Figure 4). Despite arguments about researchers' goals and skills determining approaches to uncertainty (Ma / Xiao 2021), our results confirm a persistent theory-practice gap (Windhager et al. 2019; Panagiotou et al. 2023), reflecting broader DH trends favouring simplistic visualisations repurposing over interpretative, critical approaches rooted in humanistic epistemologies (Allington et al. 2016; Windhager et al. 2019; Viola 2023).



Distribution of critical adaptations and uncertainty representation methods.

Acknowledgements

Project funded by the European Union – NextGenerationEU under the National Recovery and Resilience Plan (NRRP), Investment I.4.1 - Borse PNRR Patrimonio Culturale.

Notes

1. https://www.aiucd.it/progetti/ .
2. https://eadh.org/projects .
3. http://dhawards.org/ .
4. https://rasbt.github.io/mlxtend/api_subpackages/mlxtend.frequent_patterns /.
5. https://storymap.knightlab.com/ .
6. https://timeline.knightlab.com/ .

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